ARTHROPLASTY · NEET-SS / FET / INI-SS

Periprosthetic infection (ICM criteria)

1. Epidemiology and pathogenesis

- **Periprosthetic joint infection (PJI)** occurs in about **1-2% of primary hip and knee arthroplasties** (higher in revisions, about 3-5%). Among the commonest causes of revision (leading cause of early TKA revision).
- **Biofilm:** bacteria adhere to the implant, produce an **extracellular polymeric matrix (glycocalyx)**; sessile bacteria are **up to 1,000 times more resistant to antibiotics** and protected from host defences. **Biofilm matures within about 3-4 weeks**: the basis for implant retention only in early infections.
- **"Race for the surface"** (Gristina): host cells vs bacteria competing to colonise the implant.
- **Organisms: coagulase-negative staphylococci (S. epidermidis) and S. aureus** (together about 50-60%), streptococci, enterococci, gram-negatives, anaerobes; **Cutibacterium acnes** (shoulder, low-grade), fungi (Candida); **polymicrobial** (about 10-15%); **culture-negative** (about 7-15%: prior antibiotics, fastidious organisms).
- **Routes:** perioperative contamination (most), **haematogenous** (late acute; dental, urinary, skin, endocarditis), contiguous spread.

2. Risk factors

- **Patient: diabetes (HbA1c over about 7.5-8%, perioperative glucose over about 200 mg/dl)**, **obesity (BMI over 40)**, **smoking**, **malnutrition (albumin under 3.5 g/dl, transferrin under 200, lymphocytes under 1,500)**, **rheumatoid arthritis/immunosuppression**, **S. aureus colonisation (nasal carriers)**, chronic kidney/liver disease, **prior joint**

surgery/infection, **intra-articular steroid injection within about 3 months**, IV drug use, anaemia, male sex, depression.

- **Surgical:** prolonged operative time, **allogeneic transfusion**, haematoma, persistent wound drainage (over about 5-7 days), revision surgery, concurrent infection elsewhere.

3. Prevention

- **Optimise** glucose, BMI, nutrition, smoking cessation (4-6 weeks), treat dental/urinary infection.
- **MRSA/MSSA screening and decolonisation** (**nasal mupirocin + chlorhexidine body washes**).
- **Prophylactic antibiotics: cefazolin within 60 minutes before incision** (vancomycin within 120 minutes if MRSA-colonised or beta-lactam allergic; clindamycin alternative); **redose** after about 4 hours or with large blood loss (over 1,500 ml); **stop within 24 hours**. Antibiotic-loaded cement in high-risk patients.
- **Skin preparation: chlorhexidine-alcohol** (superior to povidone-iodine in surgical site infection trials), hair clipping (not shaving) just before surgery.
- **Theatre: laminar flow**, restricted traffic, **dilute povidone-iodine lavage** before closure (debated benefit), meticulous haemostasis, tranexamic acid (fewer transfusions).

4. Classification by timing (management-oriented)

Type	Timing	Features	Typical organism
Early post-operative (acute)	**Within about 4 weeks (up to 3 months in some systems)** of surgery	Wound drainage, erythema, fever, pain	S. aureus, gram-negatives

Type	Timing	Features	Typical organism
Acute haematogenous	Any time, **symptoms under about 3 weeks** in a previously well-functioning joint	Sudden pain, effusion, sepsis	S. aureus, streptococci
Chronic (late)	**Over about 4 weeks**/insidious	Persistent pain, early loosening, sinus tract	CoNS, C. acnes
Positive intraoperative cultures	Unexpected positive cultures at revision for "aseptic" failure	**Two or more positive**: treat as PJI (about 6 weeks)	CoNS, C. acnes

- **Tsukayama classification:** I positive intraoperative cultures, II early post-operative, III acute haematogenous, IV late chronic.

5. Diagnosis

Clinical

- **Pain** (commonest symptom), stiffness, effusion, wound problems, **sinus tract** (pathognomonic), fever (often absent in chronic infection), **early loosening** (within 5 years).

2018 ICM (International Consensus Meeting / Parvizi) definition

- **Major criteria (either = infected):**
  - **Two positive cultures of the same organism** (by standard culture methods).
  - **A sinus tract communicating with the joint or visualisation of the prosthesis.**
- **Minor criteria (pre-operative scoring):**

Criterion	Score
Serum CRP over 1 mg/dL (10 mg/L) or **D-dimer over 860 ng/mL**	**2**
Serum ESR over 30 mm/h	**1**
Synovial WBC over 3,000 cells/μL or **positive leukocyte esterase (++)**	**3**
Positive alpha-defensin (signal-to-cutoff ratio over 1)	**3**
Synovial PMN over 80%	**2**
Synovial CRP over 6.9 mg/L	**1**

    - **Total 6 or more = infected; 2-5 = possibly infected** (add intraoperative findings); **0-1 = not infected**.

- **Intraoperative findings** (for inconclusive pre-operative scores): **positive histology (over 5 neutrophils per high-power field in 5 or more fields: 3 points)**, **purulence (3 points)**, **single positive culture (2 points)**; combined score **6 or more = infected**, 4-5 inconclusive, 3 or less not infected.
- **Thresholds differ for the early post-operative period** (within 6 weeks): **CRP over 100 mg/L (knee/hip acute)**, synovial WBC over about **10,000**, PMN over about **90%** (inflammation from surgery raises markers).
- **Inflammatory arthritis, crystal disease, metallosis (ALTR)** can give false-positive synovial counts (manual count in metallosis).
- **EBJIS 2021 definition** (European Bone and Joint Infection Society) uses "infection likely/confirmed" tiers with similar markers.

Practical work-up (AAOS/ICM)

1. **Serum ESR and CRP** in every painful arthroplasty (high sensitivity: if both normal, PJI unlikely).
2. If raised (or high suspicion): **joint aspiration** (stop antibiotics at least **2 weeks** before aspiration) for **synovial WBC, PMN%, culture**

(aerobic + anaerobic; **hold cultures 10-14 days** if C. acnes or low-virulence organisms suspected), **leukocyte esterase**, **alpha-defensin**.

3. **Intraoperative: 5 (3-6) separate tissue samples** with separate instruments, **frozen section**, **sonication** of explants (improves culture yield, especially after antibiotics), avoid swabs.
4. **Imaging:** radiographs (early loosening, periosteal reaction, osteolysis); nuclear imaging (labelled leukocyte with marrow scan) when aspiration is inconclusive; MRI with MARS sequences.

6. Treatment

Debridement, antibiotics and implant retention (DAIR)

- **Indications: early post-operative (within about 4 weeks)** or **acute haematogenous (symptoms under 3 weeks)** infection, **well-fixed implants**, **healthy soft tissue (no sinus)**, susceptible organism.
- **Technique:** radical **synovectomy and debridement**, high-volume lavage (dilute povidone-iodine, chlorhexidine), **exchange of all modular parts** (head and liner/polyethylene insert: essential), followed by **pathogen-specific antibiotics** (**rifampicin** added for staphylococci with retained implants: active against biofilm; never as monotherapy; **fluoroquinolones for gram-negatives**) for about **3 months (hip) to 6 months (knee)**.
- **Success about 50-80%**; worse with **S. aureus/MRSA**, delays, high CRP, rheumatoid arthritis, prior DAIR. **Repeated DAIR** has poor results. **KLIC and CRIME80 scores** predict DAIR failure.

One-stage exchange

- Remove all components, radical debridement, re-implant new components (often **antibiotic-loaded cement**), specific antibiotics. Increasingly used for **chronic infection** with **known organism and susceptibilities**, healthy soft tissues, no sinus (debated),

immunocompetent host (Frankfurt/ENDO-Klinik practice). Similar control rates to two-stage in selected cases, fewer operations.

Two-stage exchange (traditional gold standard for chronic PJI, especially in North America)

1. **Stage 1:** remove all components and cement, radical debridement, **antibiotic-loaded cement spacer** (static or **articulating**; high-dose antibiotics: e.g. **vancomycin + gentamicin/tobramycin**; heat-stable antibiotics, about **3-4 g of antibiotic per 40 g cement**), followed by **4-6 weeks of IV/oral antibiotics**.
2. **Antibiotic holiday** (about 2 weeks; debated) and assessment of infection control (**ESR/CRP trends, aspiration**: no single test reliably confirms eradication).
3. **Stage 2:** re-implantation (often after **6-12 weeks**), intraoperative cultures/frozen section.
4. **Success about 85-90%**. **Interim mortality and failure to reach stage 2** are significant in elderly patients.
5. **After re-implantation:** a further **3 months of oral antibiotics** reduced failure in one RCT (Frank et al.).

Salvage

- **Resection arthroplasty (Girdlestone for the hip)**, **knee arthrodesis** (intramedullary nail or external fixation), **amputation** (uncontrolled infection, life-threatening sepsis, massive bone/soft-tissue loss), **long-term suppressive antibiotics** (unfit patients or retained infected implants).

Antibiotic principles

- Bactericidal, biofilm-active agents with good bone penetration: **rifampicin (staphylococci)**, **fluoroquinolones (gram-negatives)**; adjust to cultures; **monitor toxicity** (rifampicin interactions, vancomycin nephrotoxicity, linezolid myelosuppression).

- **Oral vs IV: OVIVA trial** (2019): **oral antibiotics were non-inferior to IV** within the first 6 weeks for bone and joint infection (including PJI), with fewer line complications.
- **Duration: DATIPO trial** (2021): **12 weeks of antibiotics** was superior to 6 weeks for PJI (DAIR/one-stage/two-stage) in persistent infection rates.

7. Special situations

- **Acute haematogenous infection with a well-fixed implant:** DAIR quickly (ideally within days of symptom onset).
- **Fungal PJI:** two-stage, antifungal-loaded spacer (amphotericin/voriconazole), prolonged antifungals; poor outcomes.
- **C. acnes** (shoulder): slow growing (hold cultures 14 days), male, young, previous shoulder surgery.
- **Antibiotic prophylaxis for dental procedures** in arthroplasty patients: **not routinely recommended** (AAOS/ADA), except selected immunocompromised patients.

Exam pearl: **2018 ICM major criteria: two positive cultures of the same organism, or a sinus tract. Minor: CRP/D-dimer (2), ESR (1), synovial WBC over 3,000 or LE ++ (3), alpha-defensin (3), PMN over 80% (2), synovial CRP (1); 6 or more = infected.**

Exam pearl: **DAIR (with modular exchange) for early post-operative (within 4 weeks) or acute haematogenous infection (under 3 weeks symptoms) with stable implants; add rifampicin for staphylococci. Chronic infection: one- or two-stage exchange.** OVIVA: oral non-inferior; DATIPO: 12 weeks beats 6.

Exam pearl: **Biofilm matures in about 3-4 weeks; CoNS and S. aureus are commonest; stop antibiotics 2 weeks before aspiration; take 5 tissue samples; sonicate implants.** Cefazolin within 60 minutes of incision; chlorhexidine-alcohol skin prep.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.